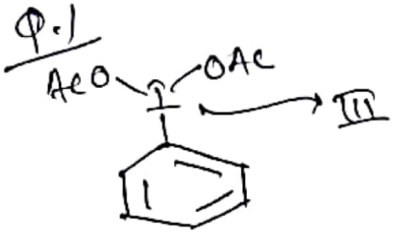
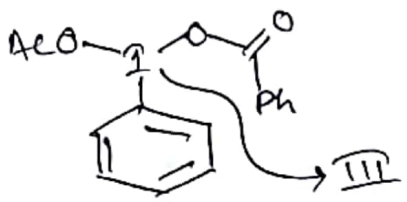
Question Set -1

Q.1



O.N = 1 + 1 + 1 = 3

due to → O O C



O.N = 1 + 1 + 1 = 3

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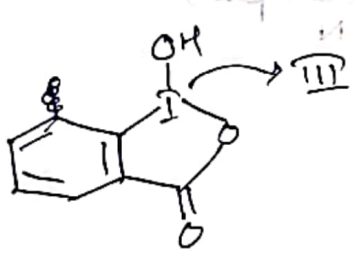
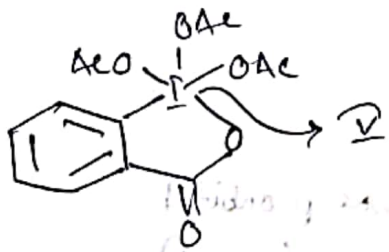


O.N = 1 + 1 + 1 = 3

due to → O O C



O.N = 1 + 1 + 1 + 1 + 1 = 5

due to → O O O O C

* Note: For each bond between I and an atom of higher e-negativity than I, it will gain an ox. number of +1. Sum of all numbers is equal to the oxidation ~~state~~ number

* Note: C is considered more electronegative than I.

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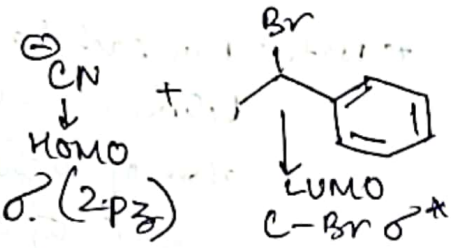
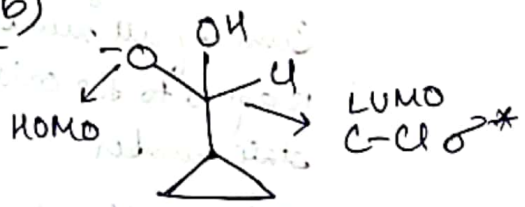
Q.2

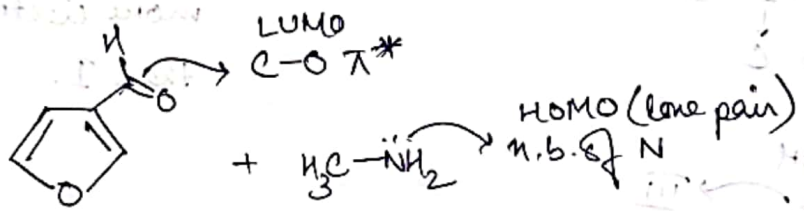
(a)



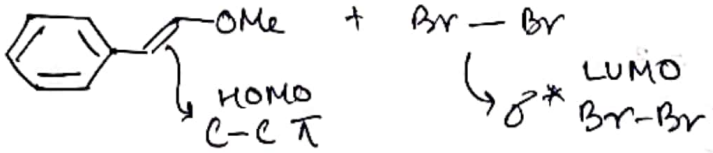
(b)



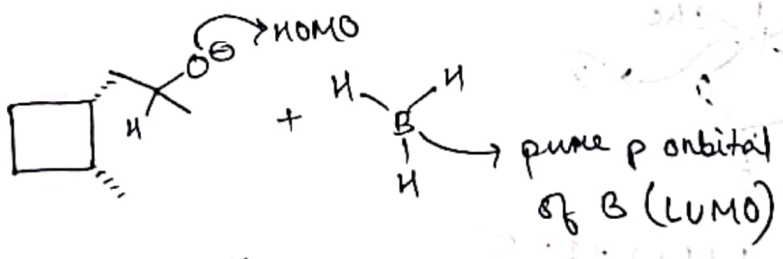
(c)



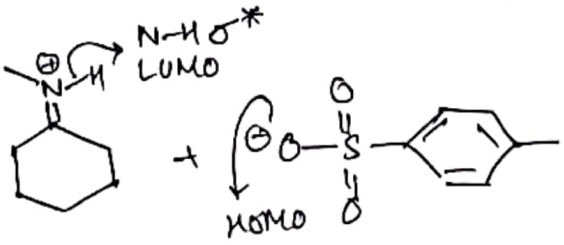
(d)



(e)



(f)



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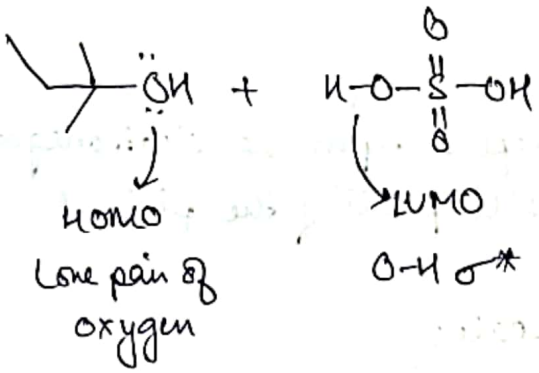
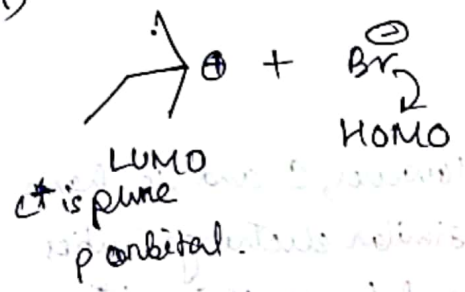
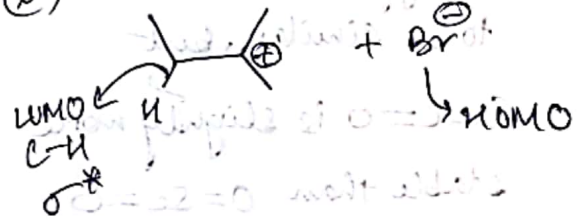
(g)



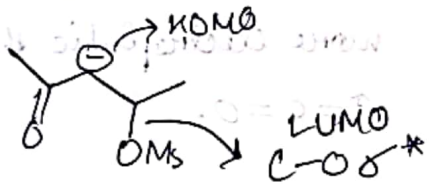
(h)



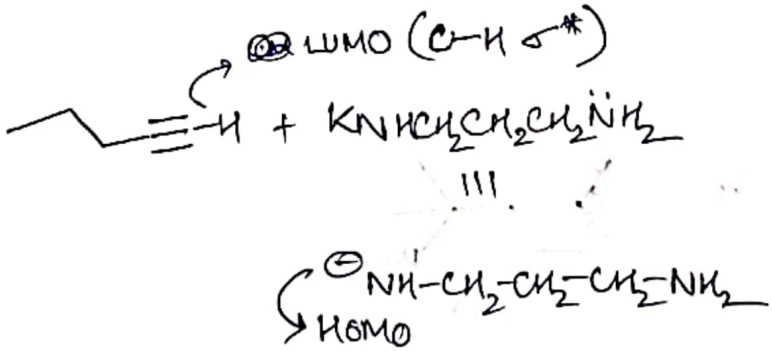
(i)



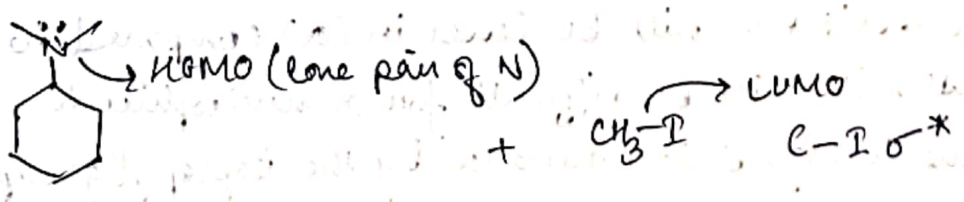
(j)



(k)



(l)



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Q.3

The π^* energy levels depend upon the electronegativity difference between the participants of the pi bond.

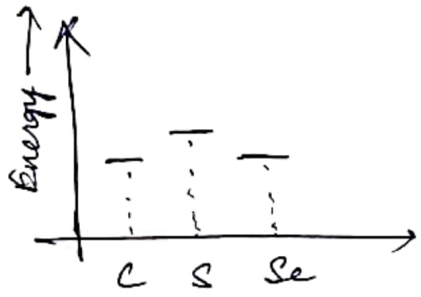
According to Pauling scale;

EN of the following are

C → 2.55

S → 2.58

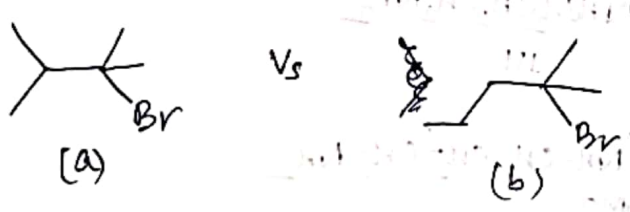
Se → 2.55



* However, C and Se have similar electronegativities and hence their pi* energy levels are expected to be similar, but O=C=O is slightly more stable than O=Se=O. Hence, O=Se=O is more electrophilic than O=C=O.

Q.4

(i)



Rate of substitution will be lesser in (a) compared to (b) because, it will be difficult for a nucleophile to attack due to high steric hindrance by the isopropyl group.

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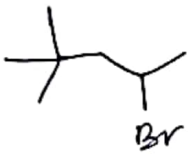
Also, after elimination we get the following products

(a)* and (b)*

clearly, (a)* is more stable than (b)* thermodynamically, and hence, (a) will undergo elimination quite easily as we are applying heat.

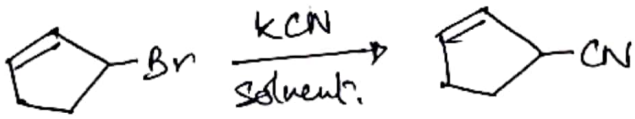
(b) <u>Omit</u>.

(c)

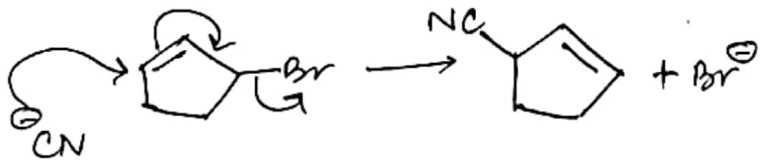


Elimination product will be more in case of bulky base. hence in presence of NaO^tBu, ^tBuOH elimination will be favoured.

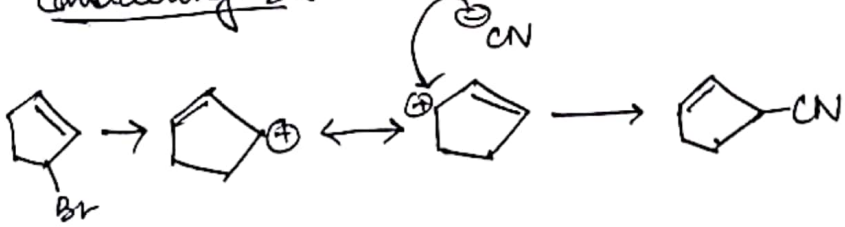
Q.5.



Considering SN^2



Considering SN^1



* In this case $^{\ominus}CN$ is a very strong nucleophile and not a very strong base. ∴ It will prefer substition to elimination. Also we are not providing any heat or alcoholic conditions.

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